

FlowRecon-Bench: A benchmark for sparse-sensor flow-field reconstruction across flow regimes, with calibrated uncertainty

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Reconstructing a full flow field from sparse point sensors is a core task of experimental and operational fluid mechanics, and a rapidly growing machine-learning literature now addresses it with deterministic operator learners and conditional generative models. Yet the field has no shared benchmark: published methods are evaluated on incompatible, often temporally leaky splits, ensembles are drawn but never scored with proper probabilistic metrics, and classical baselines are rarely reported. We present FlowRecon-Bench, a benchmark spanning five flow regimes — laminar cylinder vortex shedding (2D, multi-Reynolds), two-dimensional Kolmogorov turbulence, three-dimensional isotropic turbulence (JHTDB), a multiphysics wildfire LES with realistic corrupted measurement operators, and surface-only sensing on a family of unstructured aircraft-wing meshes — evaluated with leakage-audited splits, frozen sensor manifests, per-channel accuracy, proper scoring rules (CRPS, coverage, spread-error), physics metrics (windowed spectra, structure functions), and matched cost accounting (training and inference runtime and peak memory) for every method. Across a fidelity-audited fleet of twelve-plus methods — interpolation and gappy-POD anchors, deterministic operator learners, latent and ambient generative models including DMF-Gen — we find that method ranking is regime-dependent: [*TODO: headline numbers once 2D results land*]. No method recovers unobserved channels of 3D turbulence; every scored ensemble is under-dispersed, and post-hoc recalibration partially repairs it. We release the datasets, splits, sensor manifests, configurations, and evaluation driver as a versioned artifact. [*TODO: finalize abstract numbers and artifact DOI*]

I. INTRODUCTION

[*TODO: Port from iclr2027 intro; recenter on: (i) the task; (ii) why no benchmark exists (audit evidence: PDEBench/The Well/CFDBench are forward-surrogate or uniform-downsampling SR; the ocean SSH challenge is the only leaderboard-backed sparse-obs analogue); (iii) the leakage cautionary tale (shuffled adjacent-frame JHU split: latent-FM 0.14 $\rightarrow$ 0.56 rel- L_2 under the honest split); (iv) contributions = benchmark artifact + audited fleet + findings.*]

II. BENCHMARK DESIGN

A. Task and datasets: five flow regimes

[*TODO: Regime-ladder table: cylinder (new, OpenFOAM, cross-Re), Kolmogorov 256² (Shu et al.), JHU 4-cube 125³ cross-cube, FireBench operator matrix, SHIFT-WING surface-only. Port dataset appendix from iclr2027 App. C; add cylinder generation details + Strouhal V&V; Kolmogorov trajectory-holdout split.*]

B. Splits and leakage control

[TODO: Spatial holdout (cube), temporal blocks with gaps, cross-Re, cross-trajectory; TUNE/TEST separation; fingerprinted eval driver.]

C. Sensor protocols and measurement operators

[TODO: Random interior points 0.1–2%; surface taps (cylinder + wing); operator families (noise, occlusion, channel dropout); per-frame/per-trajectory/global taxonomy language.]

D. Metrics

[TODO: Per-channel $\text{rel-}L_2$ (observed vs unobserved); CRPS, coverage, spread-error, rank histograms; windowed shell spectra (document the leakage pitfall), structure functions; cost: train s/step, GPU-h to finished model, peak train memory, inference s/field, peak inference memory — same harness for every method.]

E. Fairness methodology

[TODO: Upstream-diff fidelity audits; parameter matching ($\sim 6.5\text{M}$); identical augmentation/sensors/normalization/estimator; budget matching; tuning-budget disclosure table.]

III. METHODS EVALUATED

[TODO: One paragraph per family: (i) classical anchors — IDW, KD-tree, gappy POD, constant; (ii) deterministic operator learners — Senseiver, FNO3D/GeoFNO, DeepONet(++); (iii) latent generative — latent flow matching, CoNFILd; (iv) ambient generative — Gen4Turb voxel diffusion, S3GM, SiT-point, and DMF-Gen [TODO: cite Wang et al.] applied at 3D scale. Equal length per method; capability matrix (which methods can run on which regimes, honest N/A rows).]

IV. RESULTS BY REGIME

A. Laminar vortex shedding: where classical methods suffice

[TODO: Cylinder table + cross-Re generalization + surface-tap variant.]

B. Two-dimensional Kolmogorov turbulence

[TODO: Kolmogorov table + operator variants + spectra/enstrophy.]

C. Three-dimensional isotropic turbulence

[TODO: Port the canonical per-channel fleet table (JHU cross-cube, $n=50$) from iclr2027 (post-P1 version); paired statistics with CIs/TOST.]

D. Multiphysics LES under corrupted operators

[TODO: Port FireBench operator matrix + robust-vs-clean ablation.]

E. Unstructured geometry with surface-only sensing

[TODO: Wing baseline table (point-native subset + N/A rows) — pending runs.]

V. CROSS-CUTTING ANALYSES

A. The distortion–perception tradeoff

[TODO: rel- L_2 rewards posterior-mean collapse; CRPS picks interior NFE; never rank generative models on rel- L_2 alone.]

B. Unobserved-channel identifiability

[TODO: Fleet-wide rel $L_2 \gtrsim 0.9$ on unobserved channels of 3D turbulence; 2D cylinder observe- u -only comparison.]

C. Calibration and post-hoc repair

[TODO: Every scored method under-dispersed; TUNE-fit spread rescaling + distance-binned conformal; pre/post coverage for every generative row.]

D. Spectral fidelity

[TODO: Windowed spectra: latent methods truncate the dissipation range $\sim 5\times$; ambient methods under-fill the inertial range; single-sample reporting rule.]

E. Computational cost and scaling

[TODO: Pareto figure; memory-vs-resolution walls (voxel diffusion OOM 320^3 , conv-AE 640^3 ; point methods flat to 1024^3).]

F. How much does leakage distort rankings?

[TODO: Same models, shuffled vs honest split.]

VI. DISCUSSION

[TODO: Practitioner decision tree (regime $\times$ sensor budget $\times$ UQ need $\times$ compute); limitations (single-GPU scope, no experimental data, tuning asymmetry disclosure); V&V framing per the special-issue call.]

VII. CONCLUSION

[TODO:]

ACKNOWLEDGMENTS

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DATA AVAILABILITY

[TODO: Artifact release (Zenodo/HF): data or regeneration recipes (JHTDB terms), sensor manifests, splits, configs, eval driver. Needs sign-off before release.]